Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Configuring Generic S3 Connectors 6.5.0

This connector allows Stellar Cyber to pull data sent to S3 and add the records to the data lake.

The Generic S3 connector uses Amazon Simple Queue Service (SQS) queues and S3 buckets to pull data. The SQS queue gets information about newly added S3 objects. Stellar Cyber reads from the queue to get information about the S3 bucket and key for the newly added files. When there is a new upload, notifications are sent to the SQS queue URL.

This is a generic S3 connector that can support a wide-range of data sources. Sample sources are included with this connector but new sources can also be added. This requires some additional information to be configured and uploaded for each new source from which you want to pull data. Contact Stellar Cyber Customer Support to assist with this.

Stellar Cyber connectors with the Collect function (collectors) may skip collecting some data when the ingestion volume is large, which potentially can lead to data loss. This can happen when the processing capacity of the collector is exceeded.

Connector Overview: Generic S3

Capabilities

- Collect: **Yes**
- Respond: **No**
- Native Alerts Mapped: **Yes*** 6.0.0
- Runs on: **DP**
- Interval: **5 minutes** (if there is no data) or, reads notifications from the SQS queue (if there is data)

Collected Data

Content Type	Index	Locating Records
N/A (collects all data in the provided SQS Queue URL)	Syslog (for CrowdStrike (JSON), Fortinet Lacework CNAPP	msg_class: `generic_s3_cloudtrail` (for

Content Type	Index	Locating Records
	(JSON), Generic (JSON), Proofpoint (JSON), Teleport (JSON), Cato Networks (JSON), and AWS Config (JSON)) AWS Events (for AWS CloudTrail (JSON)) Assets (for Fortinet Lacework CNAPP (JSON) machine_details file type) Traffic (for Cato Networks (JSON)) IDPS/Malware Sandbox Events (for Cato Networks (JSON))	AWS CloudTrail (JSON)) `generic_s3_crowdstrike_log` (for CrowdStrike (JSON)) `generic_s3_fortinet_lacework_<file_type>` for example, `generic_s3_fortinet_lacework_alert_details` (for Fortinet Lacework CNAPP (JSON)) 6.0.0 `generic_s3_log` (for Generic (JSON)) `generic_s3_proofpoint_log` (for Proofpoint (JSON)) `generic_s3_teleport_log` (for Teleport (JSON)) `cato_generic_s3` (for Cato Networks (JSON)) `generic_s3_aws_config` (for AWS Config (JSON)) 6.5.0 msg_origin.source: `s3` (for AWS CloudTrail (JSON), CrowdStrike (JSON), Fortinet Lacework CNAPP (JSON), Generic (JSON), Proofpoint (JSON), Teleport (JSON), and AWS Config (JSON)) `cato_networks` (for Cato Networks (JSON)) msg_origin.vendor: `aws` (for AWS CloudTrail (JSON), Generic (JSON), and AWS Config (JSON)) `crowdstrike` (for CrowdStrike (JSON)) `fortinet` (for Fortinet Lacework

Content Type	Index	Locating Records
		CNAPP (JSON)) 6.0.0 `proofpoint` (for Proofpoint (JSON)) `teleport` (for Teleport (JSON)) `cato_networks` (for Cato Networks (JSON)) msg_origin.category: `paas` (for AWS CloudTrail (JSON), CrowdStrike (JSON), Generic (JSON), Proofpoint (JSON), Teleport (JSON), and AWS Config (JSON)) `cloudsec` (for Fortinet Lacework CNAPP (JSON)) 6.0.0 `sase` (for Cato Networks (JSON))

Domain

<SQS Queue URL>

where <SQS Queue URL> is a variable from the configuration of this connector

Response Actions

N/A

Third Party Native Alert Integration Details 6.0.0

*For Fortinet Lacework

This connector pulls data sent to S3 for Fortinet Lacework to get the raw alerts that are stored in the Syslog index.

Stellar Cyber maps Fortinet Lacework alerts. The alerts are read from the Syslog index, enriched with Stellar Cyber fields, and mapped (with deduplication) to the Alerts index.

Deduplication is by `tenantid` and `fortinet_lacework.EVENT_ID`.

For details, see Integration of Third Party Native Alerts.

Required Credentials and Configurations

- For Access Key / Secret Key Auth Method: Access Key and Secret Key, plus Region, SQS Queue URL, File Type, and Log Source
- For IAM Role Auth Method: Source Account Access Key, Source Account Secret Key, and IAM Role ARN, plus Region, SQS Queue URL, File Type, and Log Source

Locating Records 6.0.0

To search the alerts in the Alerts index or the Original Records in the Syslog index, use the query: `msg_class: generic_s3_fortinet_lacework_alert_details AND _exists_: event.threat.name`

Adding a Generic S3 Connector

To add a Generic S3 connector:

1. Obtain credentials
2. (Optional) Creating an IAM Role Auth Method
3. Configure S3 event notifications
4. Add the connector in Stellar Cyber
5. Test the connector
6. Verify ingestion

Obtaining Credentials

To obtain the required credentials, refer to the following sections:

- Permissions
- Prerequisites
- Configuring a Bucket for Notifications (SQS Queue)
- Adding an AWS User for Stellar Cyber
- Obtaining the Access Key and Secret Key

Permissions

You need the following permissions to create the SQS queue:

- s3:GetBucketNotification
- s3:PutBucketNotification
- sqs:CreateQueue
- sqs:DeleteQueue
- sqs:GetQueueAttributes
- sqs:GetQueueUrl
- sqs:SetQueueAttributes

You also need the following permissions to perform actions:

- s3:GetObject
- s3:ListBucket
- iam:CreateRole
- iam:DeleteRolePolicy
- iam:GetRole

- iam:PutRolePolicy

Prerequisites

The prerequisite for the following procedure is that you are already sending data to an S3 bucket.

The S3 bucket is limited to data from only one log source.

Configuring the Bucket for Notifications (SQS Queue)

The procedures in this document are for an S3 bucket and SQS queue that are created in the same AWS account.

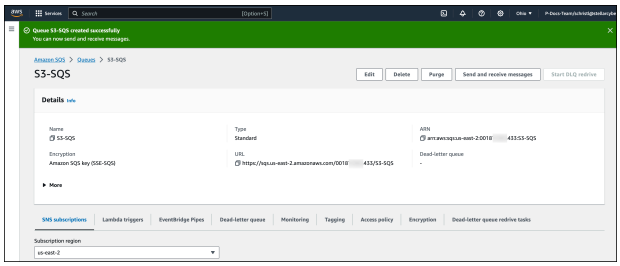
Have the following information handy:

- AWS account ID
- S3 bucket name

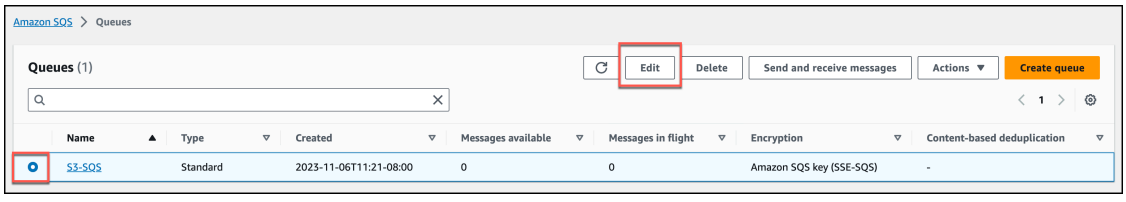
In this procedure you create an SQS queue, attach an access policy to the queue, and add a notification configuration to the bucket.

Configure only one SQS queue URL per S3 bucket.

1. Log in to the Amazon SQS console at https://console.aws.amazon.com/sqs/.
2. In the navigation pane, choose **Queues**.
3. Click **Create queue**.
    a. For **Type**, the **Standard** queue type is set by default.
    b. Enter a **Name** for your queue.
    c. For **Server-side encryption**, **Enabled** is set by default.
    d. For **Encryption key type**, **Amazon SQS key (SSE-SQS)** is set by default.
    e. Click **Create queue**.



    f. Note the **URL**. This is the **SQS Queue URL**, which you will need when configuring the connector in Stellar Cyber.
    g. Note the **ARN**. This is the SQS-queue-ARN, which you will need when configuring the access policy in the following step.
4. Replace the access policy attached to the queue:
    a. In the Amazon SQS console, in the **Queues** list, select the queue name.



b. Click **Edit**.

c. Replace the ss policy attached to the queue. Provide the following:

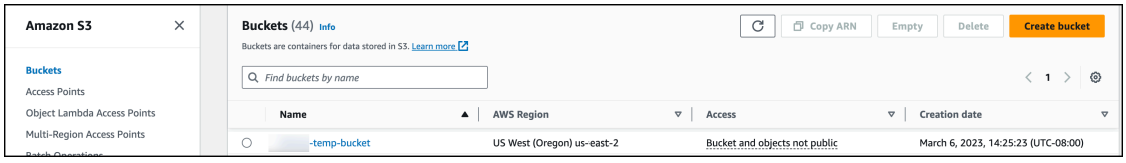
- Amazon SQS ARN you noted above
- your AWS account ID for <bucket-owner-account-id>
- your source S3 bucket name for <your-S3-bucket-name>

```
{
  "Version": "2012-10-17",
  "Id": "example-ID",
  "Statement": [
    {
      "Sid": "example-statement-ID",
      "Effect": "Allow",
      "Principal": {
        "Service": "s3.amazonaws.com"
      },
      "Action": "SQS:SendMessage",
      "Resource": "arn:aws:sqs:<aws-region:account-id:queue-name>",
      "Condition": {
        "StringEquals": {
          "aws:SourceAccount": "<bucket-owner-account-id>"
        },
        "ArnLike": {
          "aws:SourceArn": "arn:aws:s3:::<your-S3-bucket-name>"
        }
      }
    }
  ]
}
```

d. Click **Save**.

5. Add a notification configuration to your bucket.

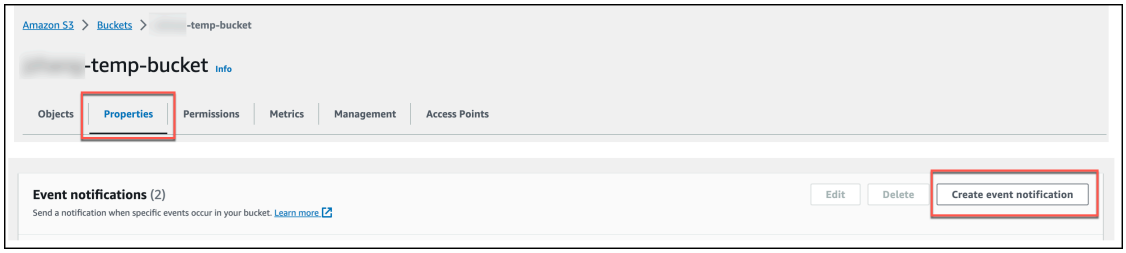
a. In the Amazon S3 Buckets console, locate your S3 bucket.



b. Double-click your bucket name.

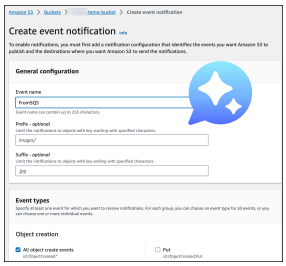
c. Click the **Properties** tab.

d. Scroll down to **Event notifications** and click **Create event notification**.



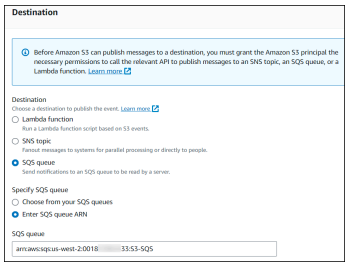
e. Enter an **Event name**.

f. For **Event types**, select **All object create events**.



g. Scroll down to **Destination**:

- For **Destination**, select **SQS queue**.
- For **Specify SQS queue**, select **Enter SQS queue ARN**.
- For **SQS queue**, enter the ARN.



h. Click **Save changes**.

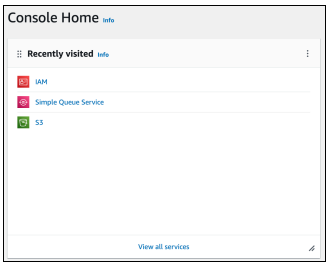
Adding an AWS User for Stellar Cyber

Create an IAM user in the same AWS account as the SQS queue. You will need the SQS-queue-ARN you noted above for the policy.

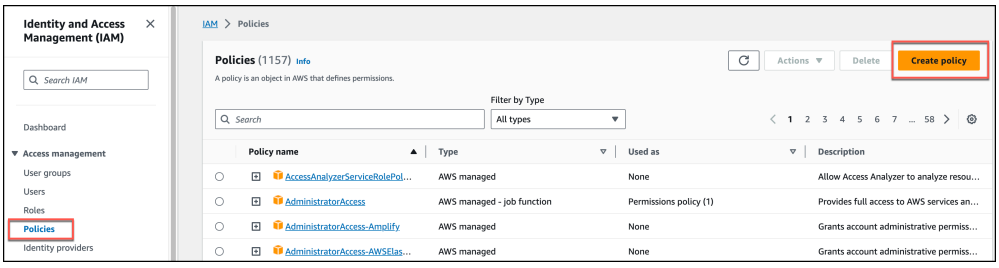
Use our example as a guideline, as you might be using a different software version.

To add a user with the appropriate permissions:

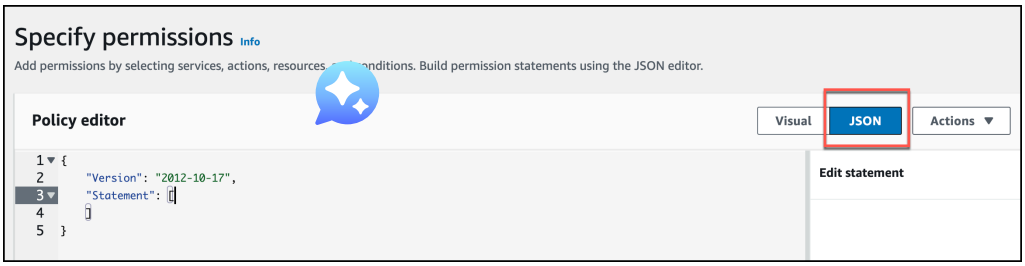
1. Log in to your **AWS Management Console** at https://aws.amazon.com/console. View the services in the Console Home or choose **View all services**.



2. Choose **IAM**. The **IAM Dashboard** appears.
3. Choose **Policies** and then choose **Create Policy**.



4. In the **Create policy** pane, choose the **JSON** tab.



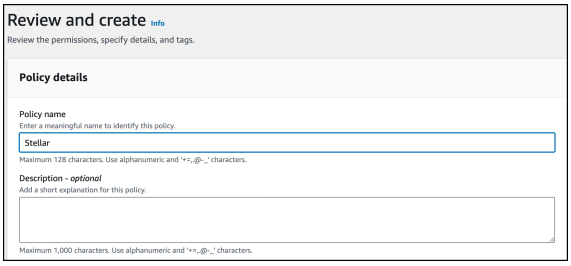
5. Using the example below as a guide, edit the JSON policy document.

> The following is just an EXAMPLE. You must modify this JSON to match the resources in your own environment.

Generic S3 policy

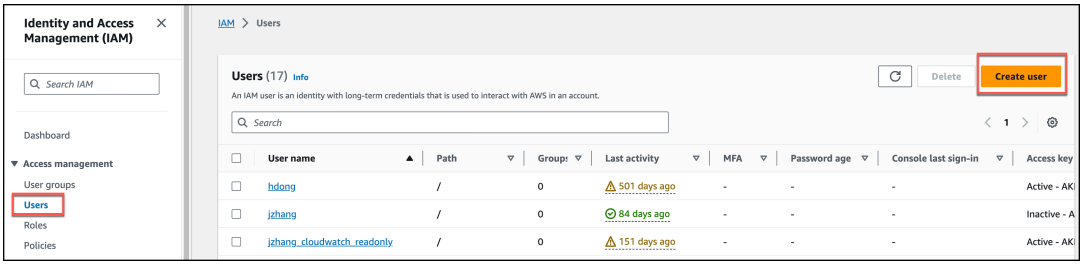
```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "sqs:DeleteMessage",
                "sqs:ReceiveMessage"
            ],
            "Resource": [
                "arn:aws:sqs:<aws-region:account-id:queue-name>"
            ]
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:GetObject",
                "s3:ListBucket"
            ],
            "Resource": [
                "arn:aws:s3:::<your-S3-bucket-name>",
                "arn:aws:s3:::<your-S3-bucket-name>/*"
            ]
        }
    ]
}
```

6. Choose **Next**.
7. Give your policy a name to associate it with Stellar Cyber, then choose **Create policy**.

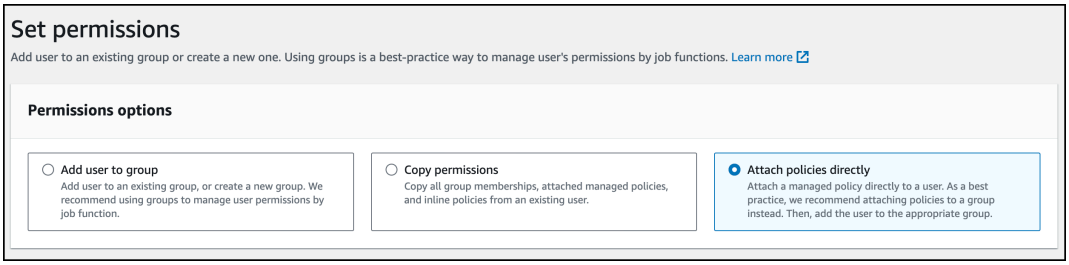


The policy can now be attached to a user.

8. From the IAM navigation pane, choose **Users** and then choose **Create user**.



9. On the **Specify user details** page for **User details**, enter a **User name** for the new user.
10. Choose **Provide user access to the – AWS Management Console** *optional* to produce login credentials for the new user, such as a password.
11. Choose how you want to create the Console password and then choose **Next**.
12. On the **Set permissions** page, choose **Attach policies directly**. Then search for the policy you created above.

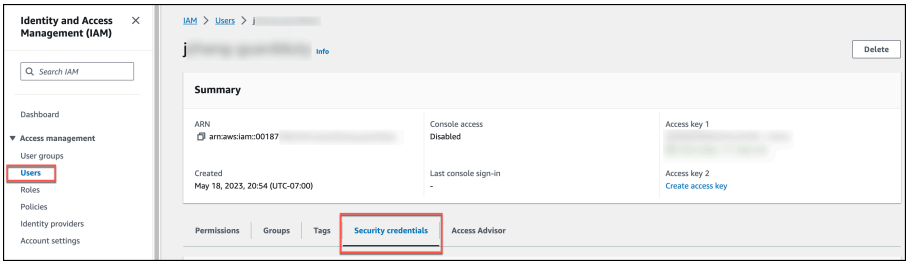


13. Select the checkbox to the left of the policy name, then choose **Next**.
14. On the **Review and create** page, verify the information and then choose **Create user**.

Obtaining the Access Key and Secret Key

To obtain the access key and secret key:

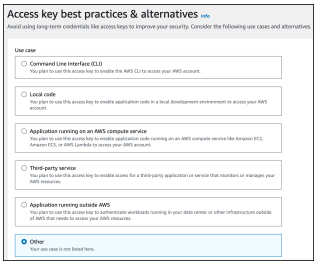
1. Log in to the IAM Console with your AWS account ID, your IAM user name, and password. You must have the permissions required to create access keys for a user.
2. Choose **Users**, then click the user name, and click **Security credentials**.



3. In the **Access keys** section, choose **Create access key**. Access keys have two parts: an access key and a secret access key. Click **Create access key**.

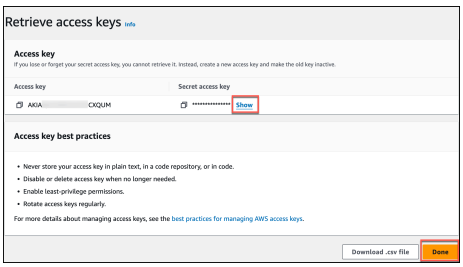


4. On the **Access key best practices & alternatives** page, choose **Other** and then choose **Next**.



5. (Optional) On the **Set description tag** page, enter a description.

6. Click **Create access key**.
7. On the **Retrieve access keys** page, choose **Show** to reveal the value of the user's secret access key. Save the access key and secret access key in a secure location. You will need them when configuring the connector in Stellar Cyber.



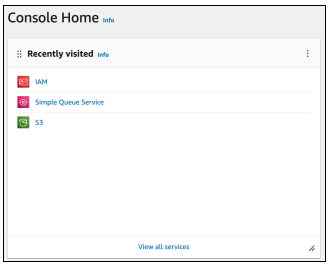
8. Click **Done**.

(Optional) Creating an IAM Role Auth Method

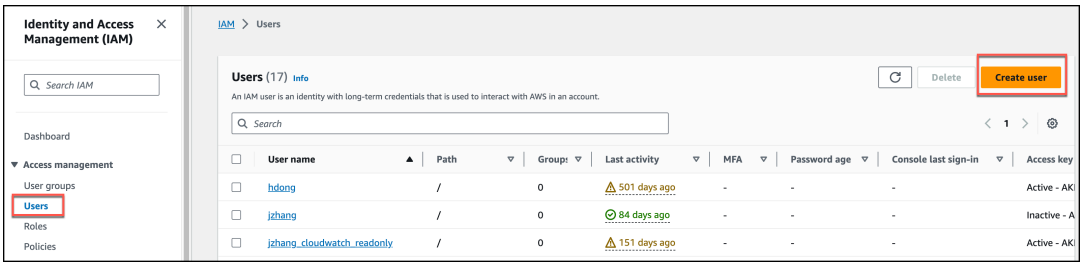
The procedures in this section are required for users who want to use the IAM Role Auth Method for the AWS resources.

Create an IAM User Without Permissions and Create IAM Policy

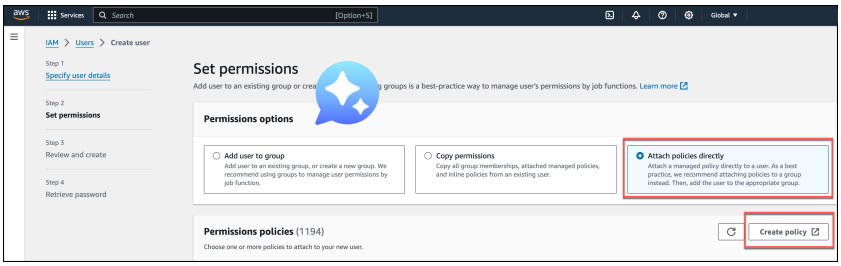
1. Log in to your **AWS Management Console** at https://aws.amazon.com/console. View the services in the Console Home or choose **View all services**.



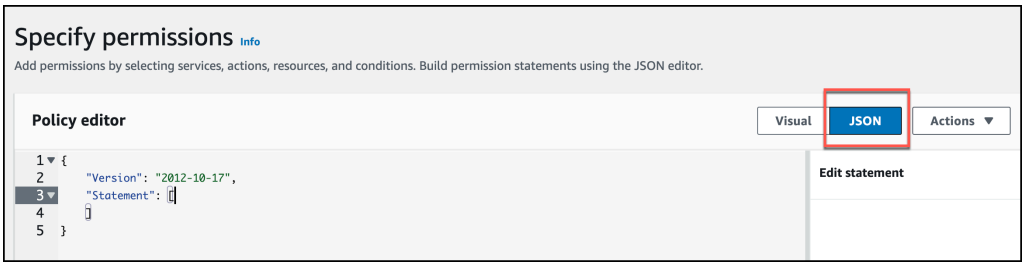
2. Choose **IAM**. The **IAM Dashboard** appears.
3. From the IAM navigation pane, choose **Users** and then click **Create user**.



4. On the **Specify user details** page for **User details**, enter a **User name** for the new user.
5. Choose **Provide user access to the – AWS Management Console** *optional* to produce login credentials for the new user, such as a password.
6. Choose how you want to create the Console password and then choose **Next**.
7. On the **Set permissions** page, choose **Attach policies directly** and click **Create policy**.



8. On the **Specify permissions** page, choose the **JSON** tab.



9. Using the example below as a guide, edit the JSON policy document.

> The following is just an EXAMPLE. You must modify this JSON to match the resources in your own environment.

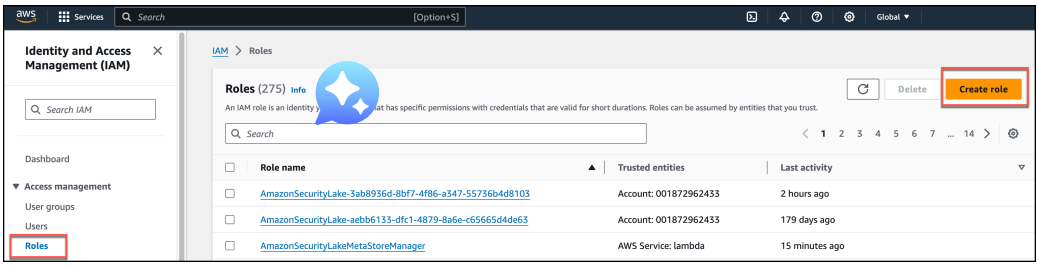
Stellar-SQS-policy

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "sqs:DeleteMessage",
                "sqs:ReceiveMessage"
            ],
            "Resource": [
                "<SQS ARN>"
            ]
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:GetObject",
                "s3:ListBucket"
            ],
            "Resource": [
                "<S3 BUCKET ARN",
                "<S3 BUCKET ARN>/*"
            ]
        }
    ]
}
```

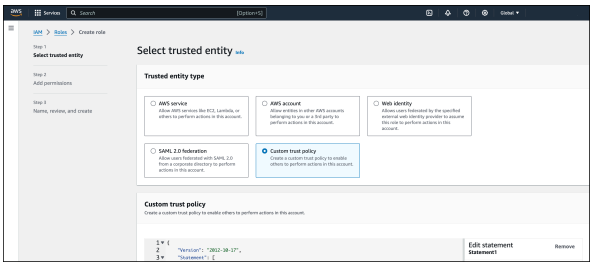
10. Give your policy a name to associate it with Stellar Cyber, for example, Stellar-SQS-policy.
11. Click **Create policy**.

Create an IAM Role with Trusted Entities

1. From the IAM navigation pane, choose **Roles** and then click **Create Role**.



2. Choose **Custom trust policy**.



3. Using the example below as a guide, edit the custom trust policy document.

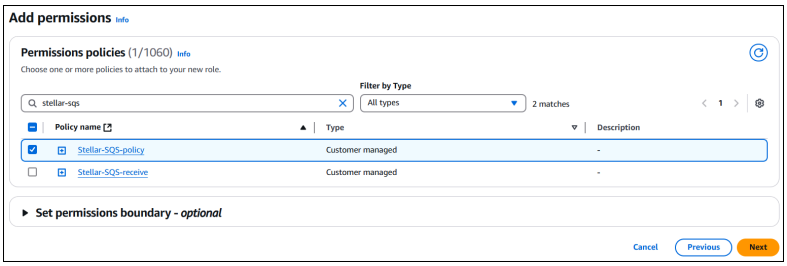
> The following is just an EXAMPLE. You must modify this JSON to match the resources in your own environment.

Trusted entities

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "Statement1",
            "Effect": "Allow",
            "Principal": {
                "AWS": "<User ARN>"
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

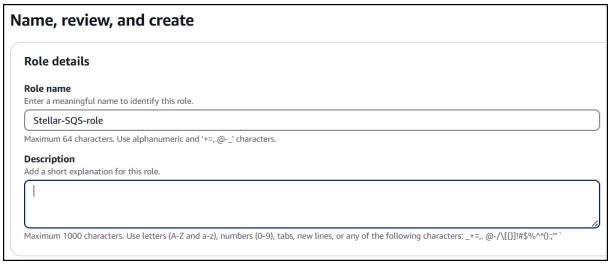
4. Click **Next**.

5. On the **Add permissions** page, select the permissions policy created earlier.

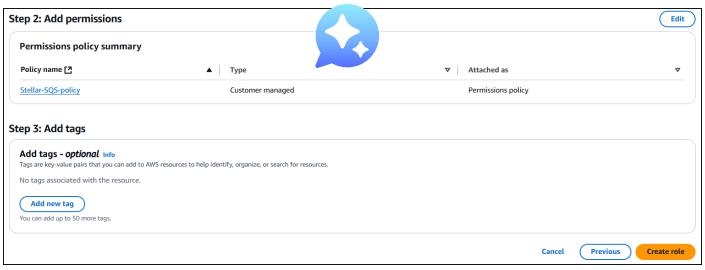


6. Click **Next**.

7. On the **Name, review, and create** page, enter a **Role name**.



8. Review the trusted entities and permissions, then click **Create Role**.



9. On the **Roles** page, use the Search box to find your role name.
10. Click your role name.
11. Copy your ARN, which you will need when configuring the connector in Stellar Cyber.

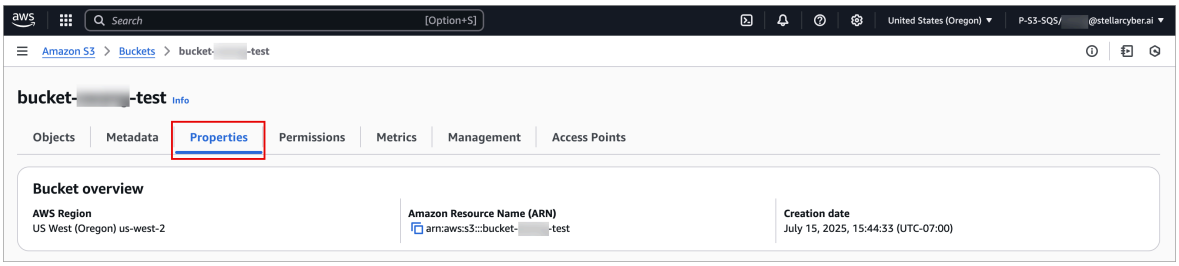
Configuring S3 Event Notifications

You can check the S3 bucket event notification configuration to ensure that only object creation events are being sent to the SQS queue. This can be done either by creating a new event notification or by modifying an existing one to include only the appropriate event types.

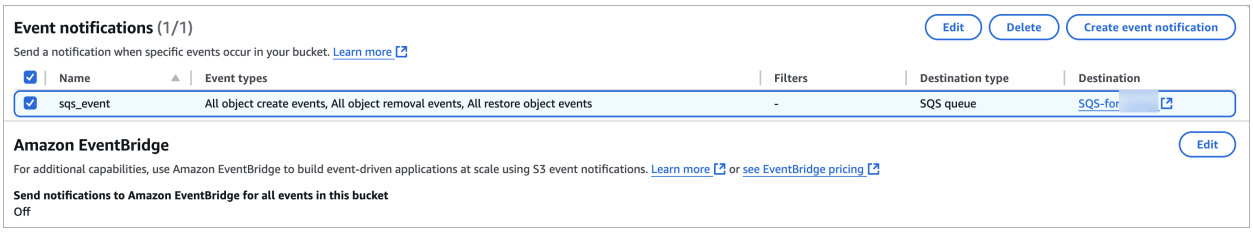
If an object delete event is sent to the SQS queue, rather than an object create event, the result could be a 404 error when you attempt to retrieve the file.

To configure only object creation events for S3 bucket event notification for the Generic S3 connector:

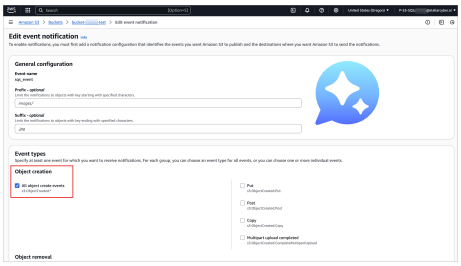
1. Go to the S3 bucket in the AWS Console.
2. Navigate to the **Properties** tab.



3. Scroll to **Event notifications**.



4. Either create a new event notification or edit an existing one.
5. For Event types, select **All object create events** only.

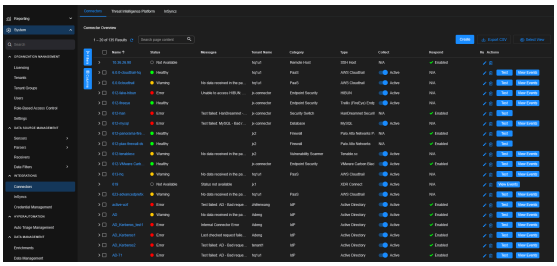


6. For Destination, choose **SQS Queue** and then select your queue.
7. Click **Save**.

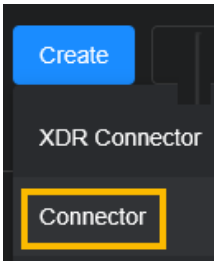
Adding the Connector in Stellar Cyber

With the access information handy, you can add a Generic S3 connector in Stellar Cyber:

1. Log in to Stellar Cyber.
2. Click **System** | **INTEGRATIONS** | **Connectors**. The **Connector Overview** appears.

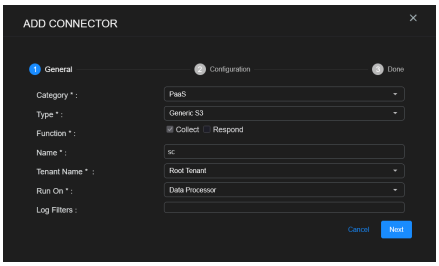


3. Click **Create** and select **Connector**.



The **General** tab of the **Add Connector** screen appears. The information on this tab cannot be changed after you add the connector.

4. Choose **PaaS** from the **Category** dropdown.
5. Choose **Generic S3** from the **Type** dropdown.



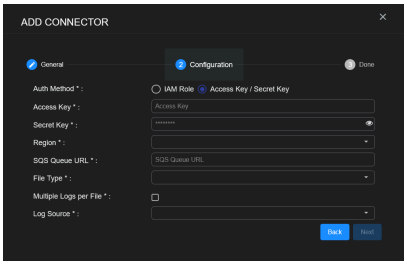
The asterisk (*) indicates a required field.

6. For this connector, the supported **Function** is **Collect**, which is enabled already.
7. Enter a **Name**.

Notes:

- This field does not accept multibyte characters.
- It is recommended that you follow a naming convention such as tenantname-connectortype.

8. Choose a **Tenant Name**. This identifies which tenant is allowed to use the connector.
9. Choose the device on which to run the connector.
10. (Optional) When the **Function** is **Collect**, you can apply **Log Filters**. For information, see Managing Log Filters.
11. Click **Next**. The **Configuration** tab appears.



The asterisk (*) indicates a required field.

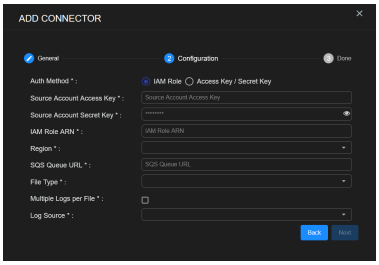
12. Choose the **Auth Method** to use **Access Key / Secret Key** or **IAM Role**.

For **Access Key / Secret Key** Auth Method:

a. Enter the **Access Key** you noted above in Obtaining the Access Key and Secret Key.

b. Enter the **Secret Key** you noted above.

For **IAM Role** Auth Method:



The asterisk (*) indicates a required field.

a. Enter the **Source Account Access Key**.

b. Enter the **Source Account Secret Key**.

c. Enter the **IAM Role ARN** you noted above in (Optional) Creating an IAM Role Auth Method.

13. Choose the **Region** from the available AWS regions in the dropdown.
14. Enter the **SQS Queue URL** you noted above. It has the following format:

https://<aws-region>.amazonaws.com/<account-id>/<queue-name>

15. Choose the **File Type**. This refers to the specific data within the S3 object. The supported file types are **gzip** and **text**. The only supported format is JSON. For example, if the **File Type** is **text**, the file must contain a properly formatted JSON object and have a .json extension.

The SQS queue should only get data in the selected format. Mixed **gzip** and raw JSON files will cause issues for the connector, regardless of the log source.

16. (Optional) If the specified **File Type** contains multiple JSON objects in the file, select the **Multiple Logs per File** checkbox. For example, if there are multiple JSON objects in the file, one per line, with each line being a single JSON object, then select this checkbox. Your file contents will look like this:

```
{ ... }
{ ... }
{ ... }
```

Do not select this checkbox if there is only a single JSON object in the file and your file contents looks like this:

```
{
   ...
}
```

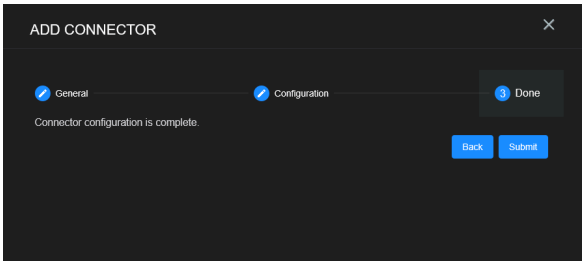
17. Choose the **Log Source**. The supported log sources are:

- **AWS Config (JSON)** 6.5.0
- **AWS CloudTrail (JSON)**
- **Cato Networks (JSON)**
- **CrowdStrike (JSON)**
- **Fortinet Lacework CNAPP (JSON)** 6.0.0 (CNAPP is the Cloud-Native Application Protection Platform)
- **Generic (JSON)**
- **Proofpoint (JSON)**
- **Teleport (JSON)**

Only one log source can be configured per connector.

In addition, there may be other Log Sources listed in this dropdown menu. Contact the Stellar Cyber Customer Support team if you have a log source you want added.

18. Click **Next**. The final confirmation tab appears.



19. Click **Submit**.

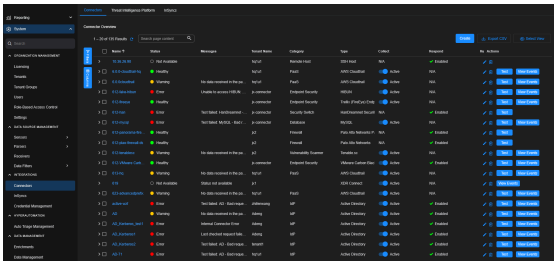
The new connector is immediately active.

Testing the Connector

When you add (or edit) a connector, we recommend that you run a test to validate the connectivity parameters you entered. The test validates authentication and connectivity.

For connectors running on a sensor, Stellar Cyber recommends that you allow 30-60 seconds for new or modified configuration details to be propagated to the sensor before performing a test.

1. Click **System | INTEGRATIONS | Connectors**. The **Connector Overview** appears.



2. Locate the connector by name that you added, or modified, or that you want to test.
3. Click **Test** at the right side of that row. The test runs immediately.

   Note that you may run only one test at a time.

Stellar Cyber conducts a basic connectivity test for the connector and reports a success or failure result. A successful test indicates that you entered all of the connector information correctly.

To aid troubleshooting your connector, the dialog remains open until you explicitly close it by using the **X** button. If the test fails, you can select the button from the same row to review and correct issues.

The connector status is updated every five (5) minutes. A successful test clears the connector status, but if issues persist, the status reverts to failed after a minute.

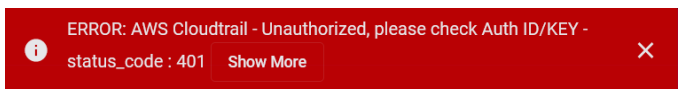
Repeat the test as needed.

Display sample messages...

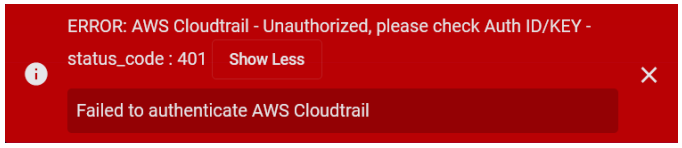
Success !



Failure with summary of issue:



Show More example detail:



If the test fails, the common HTTP status error codes are as follows:

HTTP Error Code	HTTP Standard Error Name	Explanation	Recommendation
400	Bad Request	This error occurs when there is an error in the connector configuration.	Did you configure the connector correctly?
401	Unauthorized	This error occurs when an authentication credential is invalid or when a user does not have sufficient privileges to access a specific API.	Did you enter your credentials correctly? Are your credentials expired? Are your credentials entitled or licensed for that specific resource?
403	Forbidden	This error occurs when the permission or scope is not correct in a valid credential.	Did you enter your credentials correctly? Do you have the required role or permissions for that credential?
404	Not Found	This error occurs when a URL path does not resolve to an entity.	Did you enter your API URL correctly?
429	Too Many Requests	This error occurs when the API server receives too much traffic or if a user's license or entitlement quota is exceeded.	The server or user license/quota will eventually recover. The connector will periodically retry the query. If this occurs unexpectedly or too often, work with your API provider to investigate the server limits, user licensing, or quotas.

For a full list of codes, refer to HTTP response status codes.

Verifying Ingestion

To verify ingestion:

1. Select **Threat Hunting**.

   The **Interflow Search** tab appears.

2. Change the **Indices** for the type of log source:

   - Change the **Indices** to **Syslog**.
   - For AWS CloudTrail (JSON) only, change the **Indices** to **AWS Events**.
   - For Fortinet Lacework CNAPP (JSON) only, change the **Indices** to **Assets** (msg_class: `generic_s3_fortinet_lacework_machine_details`).
   - For Cato Networks only, change the **Indices** to **Traffic**.
   - For Cato Networks only, change the **Indices** to **IDPS/Malware Sandbox Events**.

   The table immediately updates to show ingested Interflow records.



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